

View Reviews

Paper ID

80

Paper Title

CODA: CASCADED ONLINE DISCONTINUITY-AWARE ALIGNMENT FOR REAL-TIME SCORE FOLLOWING

Track Name

Papers

Reviewer #1

Questions

2. I am an expert on the topic of the paper.

Disagree

3. The title and abstract reflect the content of the paper.

Strongly agree

4. The paper discusses, cites and compares with all relevant related work

Strongly agree

5. Please justify the previous choice (Required if “Strongly Disagree” or “Disagree” is chosen, otherwise write "n/a")

n/a

6. Readability and paper organization: The writing and language are clear and structured in a logical manner.

Strongly agree

7. The paper adheres to ISMIR 2026 submission guidelines (uses the ISMIR 2026 template, has at most 6 pages of technical content followed by “n” pages of references, AI usage declaration or ethical considerations, references are well formatted). If you selected “No”, please explain the issue in your comments.

Yes

8. Relevance of the topic to ISMIR: The topic of the paper is relevant to the ISMIR community. Note that submissions of novel music-related topics, tasks, and applications are highly encouraged. If you think that the paper has merit but does not exactly match the topics of ISMIR, please do not simply reject the paper but instead communicate this to the Program Committee Chairs. Please do not penalize the paper when the proposed method can also be applied to non-music domains if it is shown to be useful in music domains.

Strongly agree

9. Scholarly/scientific quality: The content is scientifically correct.

Strongly agree

10. Please justify the previous choice (Required if "Strongly Disagree" or "Disagree" is chosen, otherwise write "n/a")

n/a

11. Novelty of the paper: The paper provides novel methods, applications, findings or results. Please do not narrowly view "novelty" as only new methods or theories. Papers proposing novel musical applications of existing methods from other research fields are considered novel at ISMIR conferences.

Agree

12. The paper provides all the necessary details or material to reproduce the results described in the paper. Keep in mind that ISMIR respects the diversity of academic disciplines, backgrounds, and approaches. Although ISMIR has a tradition of publishing open datasets and open-source projects to enhance the scientific reproducibility, ISMIR accepts submissions using proprietary datasets and implementations that are not sharable. Please do not simply reject the paper when proprietary datasets or implementations are used.

Agree

13. Pioneering proposals: This paper proposes a novel topic, task or application. Since this is intended to encourage brave new ideas and challenges, papers rated "Strongly Agree" and "Agree" can be highlighted, but please do not penalize papers rated "Disagree" or "Strongly Disagree". Keep in mind that it is often difficult to provide baseline comparisons for novel topics, tasks, or applications. If you think that the novelty is high but the evaluation is weak, please do not simply reject the paper but carefully assess the value of the paper for the community.

Disagree (Standard topic, task, or application)

14. Reusable insights: The paper provides reusable insights (i.e. the capacity to gain an accurate and deep understanding). Such insights may go beyond the scope of the paper, domain or application, in order to build up consistent knowledge across the MIR community.

Agree

15. Please explain your assessment of reusable insights in the paper.

The paper shows that leveraging the cascaded structure of score images improves score-following performance which can inspire score-following literature moving forward.

16. AI Usage Policy: The paper complies with the ISMIR 2026 AI Usage Policy for Authors. (Authors are welcome to use any tool they wish for preparing and writing the paper but they must ensure that all content is correct and original. Authors must declare any use of LLMs and AI tools if it's a part of core methodology (including

literature review, related work, generated figures, tables or other illustrations). Using AI tools for editing purposes (e.g., checking grammar and fixing typos) need not be declared. Please pay attention to any hallucinated text, figures, or references and flag them in the comments below)

Agree

17. Please justify the previous choice (Required if "Disagree" is chosen, otherwise write "n/a")

n/a

18. Write ONE line (in your own words) with the main take-home message from the paper.

The paper introduces a real-time score following system that employs a cascaded score following mechanism and silence-based jump handling, achieving state-of-the-art tracking accuracy and jump recovery.

21. Potential to generate discourse: The paper will generate discourse at the ISMIR conference or have a large influence/impact on the future of the ISMIR community.

Agree

22. Overall evaluation: Keep in mind that minor flaws can be corrected, and should not be a reason to reject a paper. Please familiarize yourself with the reviewer guidelines at <https://ismir.net/reviewer-guidelines>

Strong accept

23. Main review and comments for the authors. Please summarize strengths and weaknesses of the paper. It is essential that you justify the reason for the overall evaluation score in detail. Keep in mind that belittling or sarcastic comments are not appropriate.

The paper demonstrates state-of-the-art results on image-based score following via the proposed CODA model. It is novel and valuable that the authors leveraged the cascaded structure of sheet music images to narrow the search space and enforce tracking consistency, and also a silence-based jump recovery system. The design choices (like the two stage training pipeline) and the ablations show a carefully designed and evaluated system.

Relevant baselines are included in the thorough evaluation setup. The paper additionally contributes a repeat/jump-aware benchmark which will be valuable for the community.

The proposed training paradigm depends on annotations of system/bar/note locations in the image, which of course limits generalizability to any real-world score image and/or introduces dependence on staff/bar detection systems. But this assumption is common in score-following literature, and not a concern for the scope of this paper.

Reviewer #2

Questions

2. I am an expert on the topic of the paper.

Disagree

3. The title and abstract reflect the content of the paper.

Strongly agree

4. The paper discusses, cites and compares with all relevant related work

Strongly agree

5. Please justify the previous choice (Required if "Strongly Disagree" or "Disagree" is chosen, otherwise write "n/a")

The literature overview seems to be comprehensive; however, as a non-expert in that topic, I cannot really be 100% safe if nothing was forgotten.

6. Readability and paper organization: The writing and language are clear and structured in a logical manner.

Agree

7. The paper adheres to ISMIR 2026 submission guidelines (uses the ISMIR 2026 template, has at most 6 pages of technical content followed by "n" pages of references, AI usage declaration or ethical considerations, references are well formatted). If you selected "No", please explain the issue in your comments.

Yes

8. Relevance of the topic to ISMIR: The topic of the paper is relevant to the ISMIR community. Note that submissions of novel music-related topics, tasks, and applications are highly encouraged. If you think that the paper has merit but does not exactly match the topics of ISMIR, please do not simply reject the paper but instead communicate this to the Program Committee Chairs. Please do not penalize the paper when the proposed method can also be applied to non-music domains if it is shown to be useful in music domains.

Strongly agree

9. Scholarly/scientific quality: The content is scientifically correct.

Strongly agree

10. Please justify the previous choice (Required if "Strongly Disagree" or "Disagree" is chosen, otherwise write "n/a")

The content seems to be correct; however, as a non-expert in that topic, I cannot really be 100% safe

11. Novelty of the paper: The paper provides novel methods, applications, findings or results. Please do not narrowly view "novelty" as only new methods or theories.

Papers proposing novel musical applications of existing methods from other research fields are considered novel at ISMIR conferences.

Agree

12. The paper provides all the necessary details or material to reproduce the results described in the paper. Keep in mind that ISMIR respects the diversity of academic disciplines, backgrounds, and approaches. Although ISMIR has a tradition of publishing open datasets and open-source projects to enhance the scientific reproducibility, ISMIR accepts submissions using proprietary datasets and implementations that are not sharable. Please do not simply reject the paper when proprietary datasets or implementations are used.

Strongly agree

13. Pioneering proposals: This paper proposes a novel topic, task or application. Since this is intended to encourage brave new ideas and challenges, papers rated "Strongly Agree" and "Agree" can be highlighted, but please do not penalize papers rated "Disagree" or "Strongly Disagree". Keep in mind that it is often difficult to provide baseline comparisons for novel topics, tasks, or applications. If you think that the novelty is high but the evaluation is weak, please do not simply reject the paper but carefully assess the value of the paper for the community.

Agree (Novel topic, task, or application)

14. Reusable insights: The paper provides reusable insights (i.e. the capacity to gain an accurate and deep understanding). Such insights may go beyond the scope of the paper, domain or application, in order to build up consistent knowledge across the MIR community.

Agree

15. Please explain your assessment of reusable insights in the paper.

I see the reusable insights mainly in the concept of hierarchical/cascaded processing, which is a conceptually nice idea and transfers to many other MIR tasks. I don't expect insights from the model architecture itself, since the many DNN layers are difficult to analyze.

16. AI Usage Policy: The paper complies with the ISMIR 2026 AI Usage Policy for Authors. (Authors are welcome to use any tool they wish for preparing and writing the paper but they must ensure that all content is correct and original. Authors must declare any use of LLMs and AI tools if it's a part of core methodology (including literature review, related work, generated figures, tables or other illustrations). Using AI tools for editing purposes (e.g., checking grammar and fixing typos) need not be declared. Please pay attention to any hallucinated text, figures, or references and flag them in the comments below)

Agree

17. Please justify the previous choice (Required if "Disagree" is chosen, otherwise write "n/a")

n/a

18. Write ONE line (in your own words) with the main take-home message from the paper.

A DNN for online sheet music image-to-audio system is proposed that predicts system, bar, and note position in a hierarchical manner.

21. Potential to generate discourse: The paper will generate discourse at the ISMIR conference or have a large influence/impact on the future of the ISMIR community.

Agree

22. Overall evaluation: Keep in mind that minor flaws can be corrected, and should not be a reason to reject a paper. Please familiarize yourself with the reviewer guidelines at <https://ismir.net/reviewer-guidelines>

Strong accept

23. Main review and comments for the authors. Please summarize strengths and weaknesses of the paper. It is essential that you justify the reason for the overall evaluation score in detail. Keep in mind that belittling or sarcastic comments are not appropriate.

This paper introduces CODA, a DNN for the online alignment of music audio and sheet music images. As a novelty, the system does a cascaded prediction (system, bar within system, note within bar), and claims to be discontinuity (jump)-aware.

Disclaimer: While I am familiar with music alignment, I am not yet familiar with audio-to-image alignment with transformer-based architectures. As such, I am reviewing this paper from the point of view of a generally interested ISMIR attendee without prior work in that field.

The paper is fluently written and gives a good introduction to the audio-to-score image alignment problem, as well as the additional difficulties that arise from the system/bar/note hierarchy and jumps in the sheet music image due to repeats, D.C., etc. The related work section seems to be comprehensive and cites a variety of work from the groups that are well-known for their work in audio-score alignment.

The method section is a bit hard to follow for the non-expert reader, but it seems to provide the necessary details. Figure 1 is a good overview, but could be made more accessible by trying to include all variable names that are used in the text (e.g., where are H_t , x_t , F , f_r ... in Fig. 1?).

One thing I am not certain about is the jump detection and recovery strategy. As written in Sec. 3.6 and illustrated in Fig. 3, the authors seem to assume a silence break before a

discontinuity; this is something I would not assume to be commonly occurring in music recordings, and the authors should add a deeper explanation/intuition for their motivation.

The experimental section is comprehensive and includes many baselines, by evaluating on the common MSMD dataset. The performance of the proposed method seems to be impressive, and the small yet understandable ablation studies offer insights w.r.t. the influence of the model's components.

Overall, this paper seems to be a valuable contribution to the score following community and is definitely a good ISMIR paper.

Reviewer #3

Questions

2. I am an expert on the topic of the paper.

Strongly agree

3. The title and abstract reflect the content of the paper.

Strongly agree

4. The paper discusses, cites and compares with all relevant related work

Strongly agree

5. Please justify the previous choice (Required if “Strongly Disagree” or “Disagree” is chosen, otherwise write "n/a")

n/a

6. Readability and paper organization: The writing and language are clear and structured in a logical manner.

Strongly agree

7. The paper adheres to ISMIR 2026 submission guidelines (uses the ISMIR 2026 template, has at most 6 pages of technical content followed by “n” pages of references, AI usage declaration or ethical considerations, references are well formatted). If you selected “No”, please explain the issue in your comments.

Yes

8. Relevance of the topic to ISMIR: The topic of the paper is relevant to the ISMIR community. Note that submissions of novel music-related topics, tasks, and applications are highly encouraged. If you think that the paper has merit but does not exactly match the topics of ISMIR, please do not simply reject the paper but instead communicate this to the Program Committee Chairs. Please do not penalize

the paper when the proposed method can also be applied to non-music domains if it is shown to be useful in music domains.

Strongly agree

9. Scholarly/scientific quality: The content is scientifically correct.

Strongly agree

10. Please justify the previous choice (Required if "Strongly Disagree" or "Disagree" is chosen, otherwise write "n/a")

n/a

11. Novelty of the paper: The paper provides novel methods, applications, findings or results. Please do not narrowly view "novelty" as only new methods or theories. Papers proposing novel musical applications of existing methods from other research fields are considered novel at ISMIR conferences.

Strongly agree

12. The paper provides all the necessary details or material to reproduce the results described in the paper. Keep in mind that ISMIR respects the diversity of academic disciplines, backgrounds, and approaches. Although ISMIR has a tradition of publishing open datasets and open-source projects to enhance the scientific reproducibility, ISMIR accepts submissions using proprietary datasets and implementations that are not sharable. Please do not simply reject the paper when proprietary datasets or implementations are used.

Strongly agree

13. Pioneering proposals: This paper proposes a novel topic, task or application. Since this is intended to encourage brave new ideas and challenges, papers rated "Strongly Agree" and "Agree" can be highlighted, but please do not penalize papers rated "Disagree" or "Strongly Disagree". Keep in mind that it is often difficult to provide baseline comparisons for novel topics, tasks, or applications. If you think that the novelty is high but the evaluation is weak, please do not simply reject the paper but carefully assess the value of the paper for the community.

Disagree (Standard topic, task, or application)

14. Reusable insights: The paper provides reusable insights (i.e. the capacity to gain an accurate and deep understanding). Such insights may go beyond the scope of the paper, domain or application, in order to build up consistent knowledge across the MIR community.

Agree

15. Please explain your assessment of reusable insights in the paper.

The insights provided in this paper show a clear derivation of model architecture from musical reasoning based on the task at hand. The evaluation and ablation studies proposed in this paper are also valuable and reusable in future works. The inclusion of a

manual annotations of repeat structures in an existing dataset is also an extremely useful resource.

16. AI Usage Policy: The paper complies with the ISMIR 2026 AI Usage Policy for Authors. (Authors are welcome to use any tool they wish for preparing and writing the paper but they must ensure that all content is correct and original. Authors must declare any use of LLMs and AI tools if it's a part of core methodology (including literature review, related work, generated figures, tables or other illustrations). Using AI tools for editing purposes (e.g., checking grammar and fixing typos) need not be declared. Please pay attention to any hallucinated text, figures, or references and flag them in the comments below)

Agree

17. Please justify the previous choice (Required if "Disagree" is chosen, otherwise write "n/a")

n/a

18. Write ONE line (in your own words) with the main take-home message from the paper.

The authors propose a novel score-image based score-following system that is also capable of handling discontinuities in performance such as skips, repeats, etc.

21. Potential to generate discourse: The paper will generate discourse at the ISMIR conference or have a large influence/impact on the future of the ISMIR community.

Strongly agree

22. Overall evaluation: Keep in mind that minor flaws can be corrected, and should not be a reason to reject a paper. Please familiarize yourself with the reviewer guidelines at <https://ismir.net/reviewer-guidelines>

Strong accept

23. Main review and comments for the authors. Please summarize strengths and weaknesses of the paper. It is essential that you justify the reason for the overall evaluation score in detail. Keep in mind that belittling or sarcastic comments are not appropriate.

This paper proposes a score-image based score following system that handles discontinuities in the performance. These discontinuities can range from musically defined repeats in the score, to performance error. The authors design the system to take advantage of the cascaded score structure, thereby ensuring consistency of prediction across note, bar and system levels.

The paper details a very thorough and clear explanation of the task at hand, by covering all the related work that was done in the task-domain as well. The proposed system architecture is very clearly defined, along with the motivation behind each part of the system. The usage of FiLM conditioning as well as cross-attention, and the final beam

search methodology are not only interesting but are also based on strong musical reasoning, and are therefore well justified.

Some of the additional highlights that stood out were:

1. The incorporation of the two-phase curriculum training to handle the downstream effects of incorrect system predictions, even though the model's system prediction accuracy is already quite good.
2. The idea to suppress transition penalties and widen attention when pauses/silences are detected.

What was unfortunate was that there wasn't much room for discussion within the paper. I would have liked to know the authors' opinion and reasoning on some of the finer details of the parameters chosen, such as the silence onset of 3 frames/150 ms, energy onset and offset thresholds, and especially some insights on why the jump recovery was higher on the repeat subset than the random subset.

Congratulations to the authors on this work!